

Bumpy Sliding: An MR System for Experiencing Sliding Down a Bumpy Cliff

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ABSTRACT

In this study, we present an interactive MR system with which a player can experience a cartoon physical scene of stabbing a knife on the wall, mainly with stimuli to hands and arms, while sliding down a bumpy cliff. We developed two devices: one is a wall device that mimics the cliff surface and moves upwards; and the other is a knife-shaped device with a retractable blade with which one can realize the physical situation of pushing it and penetrating the wall surface. The speed of the sliding is controlled using the pressure data obtained at the end of the knife blade and the impact of hitting a rock with a knife is expressed using linear movement of a push solenoid attached in the handle of the knife.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices

1 INTRODUCTION

Research and development of mixed reality (MR), which combines real-world objects and environmental information with virtual setting, is recently advanced drastically. In order to make an MR experience with sense of presence, consistency between virtual setting and geometric and physical conditions of objects in the real world needs to be maintained. Many systems and contents have been developed that enable real-time experiences using products such as HoloLens and Meta Quest 2. There are also systems that have been developed to make us feel physical quantities such as weight and force in grasping or touching an object [1]. There is also a system that allows users to experience unrealistic shape change such as scaling of the body [2]. However, many of the MR representations of more fictitious and fantastical phenomena that have been expressed in novels, cartoon, animation, movies, etc. have not yet been realized.

We focus on phenomena of cartoon physics, which is a summary of laws that people accept as entertainment expressions though they are not correct in the rigorous physical sense. Many such phenomena have been accumulated and used in cartoon and animation to move the emotion of readers and viewers. The main motivation of our study is to make it possible to physically experience fictitious events in cartoon physics, which we consider is promising in the coming Metaverse era as a new way of appreciating entertainment contents. So far, we have focused on a thrilling scene in which a player, for some reason, slides down a cliff surface and tries to stab a knife into the slope to stop and survive. We developed a first version of the knife and wall devices for this purpose and presented a system as a poster [3]. In our idea, instead of making a player slide down, a belt conveyor on the wall device moves upwards on which a knife device with a



Figure 1: Image of experiencing Bumpy Sliding.

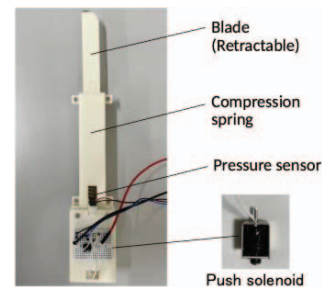


Figure 2: Knife device.

retractable blade is pressed, resulting in making the player to relatively feel the sliding down. However, the former system only targets the flat surface, and there also remain problems such as the safety on touching with the conveyor and the dependency on the stiffness of the conveyor belt.

In this study, in order to make the sliding experience more expressive, we allow the surface to be bumpy with obstacles such as rocks by embedding a solenoid in the knife device. By pushing out the iron bar inside it, the impact of being flipped by contact is presented. We developed the whole system and confirmed the feasibility of the sliding experience while enhancing the safety of the device.

2 SYSTEM

2.1 Knife Device

The knife device is not a normal type with a fixed blade, but the blade retracts into the handle when pressed to a surface. Inside the handle, a compression spring is connected to the blade, and a pressure sensor and a push solenoid (Takaha, CB1250) are installed at the bottom of the handle. A controller of Meta Quest is attached to detect the relative location to the wall device. When the knife hits some object such as a rock during the sliding, the iron bar inside the solenoid is pushed out, allowing the player to experience the change in shape and impact of a bumpy surface.

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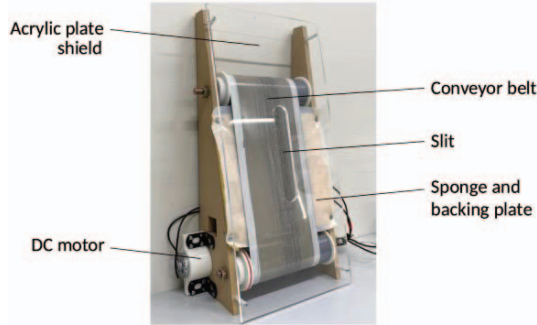


Figure 3: Wall device.

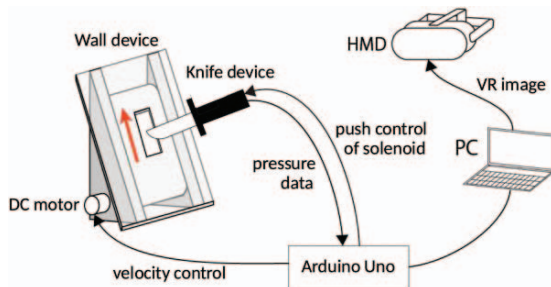


Figure 4: System overview.

2.2 Wall Device

As shown in Fig. 3, the wall device consists of a conveyor belt, a DC motor (12V-4.9kg/cm), a sponge and a backing plate behind the belt, and an acrylic plate shield to prevent hands from getting caught. The slope of the face of the wall is about 80 degrees. There is a slit in the center of the shield into which a player inserts the knife device. The belt rotates upwards when a player presses the knife on it, which gives the hand and arm a kinesthetic sensation of relatively sliding down against the wall. A Meta Quest controller is attached to the top of the wall device to track the relative position to the knife device.

The whole system is summarized in Fig. 4.

3 APPLICATIONS

The Arduino application obtains the pressure data of the knife device and controls the rotation of the motor of the wall device and the movement of the solenoid. It also sends the pressure data to PC for utilizing it to generate VR images.

The VR application was created using Unity 2020.3.10f1. Based on the pressure data of the knife device, the speed of the wall presented by the HMD, that is, the player's sliding speed, is determined. Based on the position of the controller of the hand, VR images of the hand and knife are generated. As a visual content, we created an animation of a villain who looks down a player on the cliff and drops him down. After this scene, a player slides down the cliff and tries to stop by stabbing the knife.

4 DEMONSTRATION

The experience the system provides is the following. A player holds the knife device in a hand and faces the wall device. Then, a villain appears on the cliff and lets the player fall down and the conveyor starts rotating (Fig. 5(a)). If a knife is pushed, the conveyor slows down; otherwise, it speeds up, resulting in deadly fall (Fig. 5(b)). When the knife hits a rock, a player gets the impact by the solenoid (Fig. 5(c)). When a knife is fully pressed, the player finally stops virtually and physically (Fig. 5(d)).

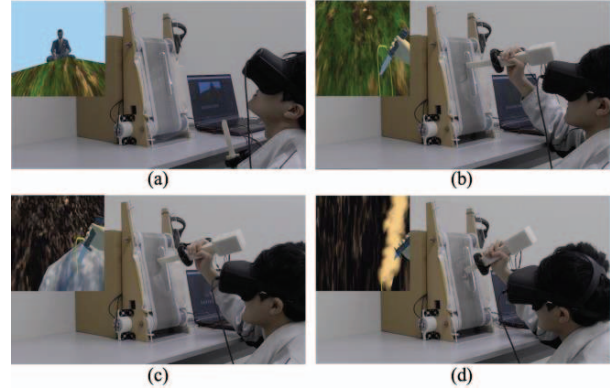


Figure 5: Interactive MR experience.

5 DISCUSSION

First of all, our system enables a player to make a multimodal experience of sliding down a cliff with stimuli, especially, around the hand and arm. Since the shield is attached on the wall for safety, the place where the knife is inserted is limited, but by making tracking and visualizing a square frame for the guide in the image, it was fully possible to insert the knife through the slit. As for the impact of the solenoid, it was not yet enough to feel the strong repulsive sensation of rocks, so we should further improve the mechanism of the device.

As applications, first we expect our system with a larger size to be used as a thrill attraction in a theme park. It is also promising to extend the system for the training to stop slipping on a snowy surface using the ice ax or pole, by modifying the knife device. Furthermore, since working with instruments on some bumpy or uneven surfaces is a rather general activity, we continue exploring further applications along this line, about which we also expect to discuss during the demonstration.

6 CONCLUDING REMARKS

We presented an MR system for experiencing sliding down a bumpy cliff using knife and wall devices. As future work, we plan to make user evaluation on the issues such as the tactile sensation when stabbing the knife device, the speed of the wall, and the sense of presence for the sliding and stopping experience. The refinement and user evaluation of the MR content should be also done. Finally, we also consider the possibility of extending the system for the whole body experience, by providing sliding sensation to the foot and by hanging the body.

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